

FARM DECISION INTELLIGENCE SYSTEM

KISAN COMPASS

Farm Decision Intelligence

"Turning fragmented farm data into decisions a farmer can actually understand."

CORE ARCHITECTURAL PHILOSOPHY

**"Not what to think.
What's actually true — and how sure we are."**

PROBLEM AREA: AI Farm Decision & Market Intelligence Agent

TEAM: [TEAM NAME] | TEAM ID: [TEAM ID] | LEAD: Darshil Nigam

SYSTEM TOPOLOGY

Integrated Farm Intelligence Loop

WEATHER ENSEMBLE

Open-Meteo 7-Day Numerical Forecast

MARKET BENCHMARKS

AGMARKNET Daily Modal Yard Prices

SOIL & AGRONOMY

ICAR Benchmark Loam • 1,845 GDD

RURAL LOGISTICS

Haversine Distance + 1.25x Detour Factor

OUTPUT: GROUNDED ACTIONABLE INTELLIGENCE

- Dynamic What-If Simulation & Downside Risk
- Traceable Net Realization (After Dedicated Freight)
- Absolute Human Veto Preserved
- Provenance Chain with Zero Fake Claims

THE FARMER DOESN'T HAVE A DATA PROBLEM.

They have a decision problem. Raw information without contextual synthesis leaves 100% of the risk on the farmer.

FRAGMENTED SIGNALS (5 SEPARATE SILOS)

• Weather Forecasts

"68% rain probability" — but does that mean harvest today, or will muddy fields trap the tractor?

• Mandi Price Bulletins

"Unnao ₹2,380 vs Kanpur ₹2,310" — but which yields more cash after ₹2,500 dedicated truck freight?

• Crop Agronomy Data

"Day 136 in ground" — but has the grain moisture dropped below 14% to avoid APMC dockage?

• Soil Moisture Telemetry

"28% field moisture" — but how many days before root-zone moisture permits heavy transport?

• Logistics & Tariffs

"28 km rural detour" — manual negotiations with local mini-truck owners with hidden rate gouging.

THE UNANSWERED DILEMMAS OF THE REAL FARMER

"Should I harvest today or hold for 4 days?"

Weighing storm damage against potential post-storm price recovery.

"Where should I sell to maximize net cash?"

Comparing higher mandi quotes against longer rural diesel freight tariffs.

"Will rain ruin grain quality and trigger dockage?"

Downside risk of APMC auction discounts for moisture-damaged grain.

"How much quantity should I commit today?"

Hedging risk by selling a fraction now and holding the remaining stand.

"Will dedicated freight eat my entire arbitrage premium?"

Navigating rural transport pricing without transparent distance calculation.

CORE INSIGHT: Information without contextual synthesis does not create decisions. Farmers don't need raw data — they need decision clarity.

MEET KISAN COMPASS

A persistent digital state of the farm, continuously combining agricultural signals into explainable decision scenarios.

CONTINUOUS MULTIMODAL CONVERGENCE

Weather Forecast
Open-Meteo NWP

Mandi Benchmarks
AGMARKNET Rates

Soil Profile
ICAR Loam Telemetry

Crop Stage
GDD Thermal Units

Rural Logistics
Haversine Detour

Farmer Style
Risk Tolerance γ

THE PERSISTENT FARM DIGITAL TWIN

- Bound to authenticated Supabase PostgreSQL cloud state (Tenant RLS)
- Farm identity: Field GPS, 7.37 Acres, Wheat HD-2967, 43.6 qtl stand
- Deterministic Net Realization Engine: (Gross Price – Dedicated Freight – Dockage)
- Simulates counterfactual horizons: Sell Now vs Wait vs Split
- Verifiable mathematical proof generated for every recommendation

WHAT THE SYSTEM ACTUALLY DELIVERS

1. Actionable Recommendation

Specific, concrete operational guidance (e.g., "Harvest now & dispatch to Unnao Mandi") rather than vague weather summaries.

2. Transparent Causal "Why"

Explicit mathematical drivers: +₹920 net arbitrage gain over closer Kanpur yard after paying ₹2,521 dedicated haulage.

3. Counterfactual Scenarios

Full simulation of alternative choices: What happens to realization if harvest is delayed +5 days into convective rainfall?

4. Explicit Uncertainty Bounds

P10/P50/P90 probability intervals instead of false single-point certainty. Clear labeling of estimation boundaries.

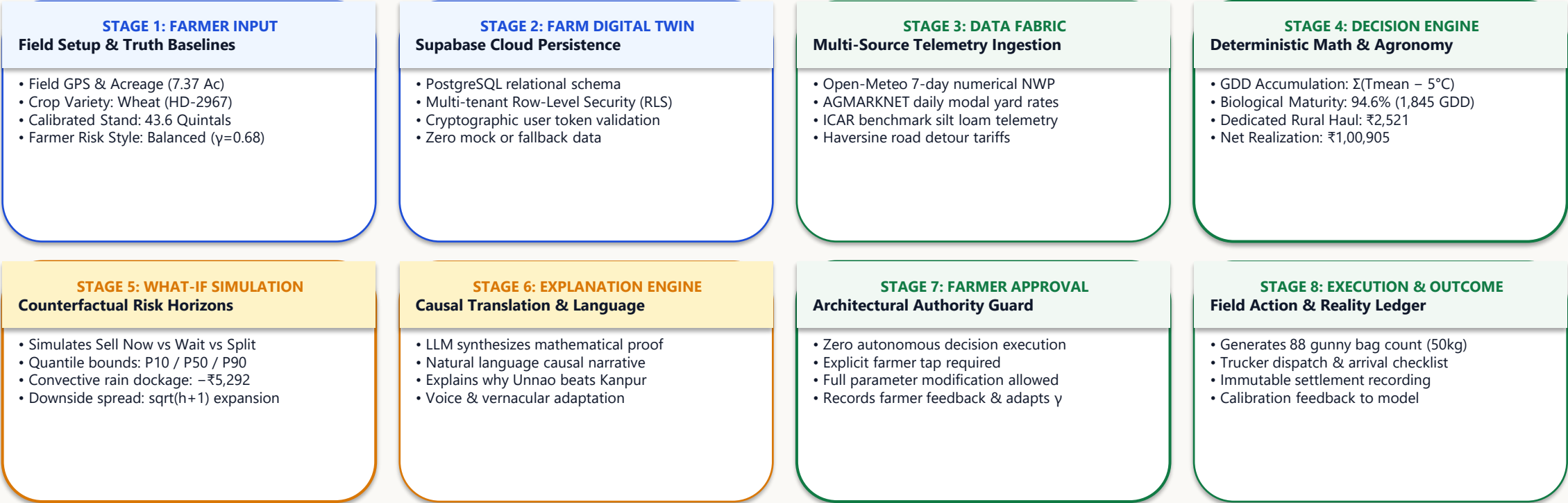
5. Absolute Farmer Authority

Human-in-the-loop checkpoint: Autonomous execution is disallowed by architecture. The farmer holds absolute veto power.

THE GOLDEN RULE: "The system recommends. The farmer decides."

FROM FIELD DATA → FARM DECISION

Engineering discipline: Deterministic computation for math and agronomy, LLMs strictly for natural language explanation.

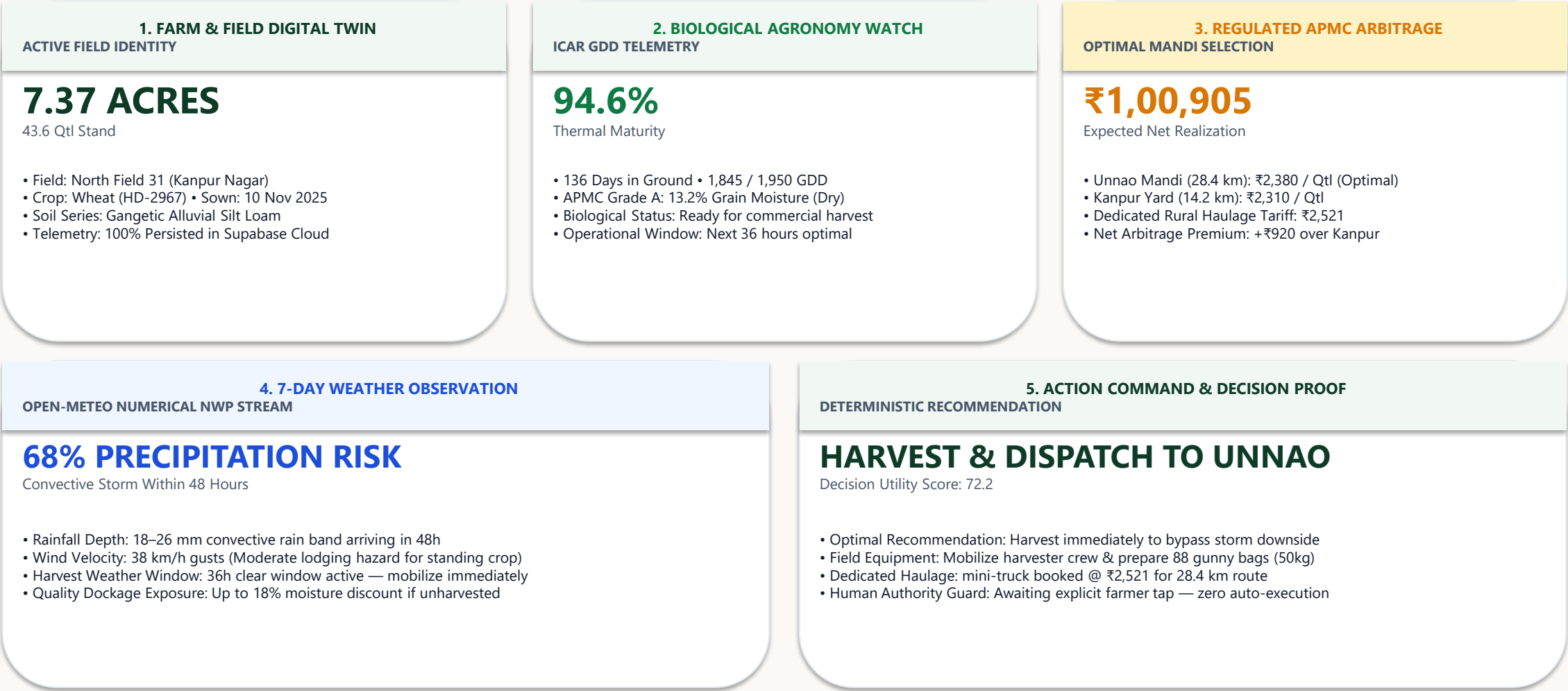


CRITICAL TECHNICAL DISCIPLINE: LLM ≠ CALCULATOR

- DETERMINISTIC CODE: All financial arithmetic, net realization, road freight tariffs, GDD agronomy, and quantile simulations are calculated via verified TypeScript & PostgreSQL code — zero mathematical hallucination.
- AI / LLM LAYER: Generative AI is deployed exclusively for translation, voice accessibility, and natural-language causal explanations.

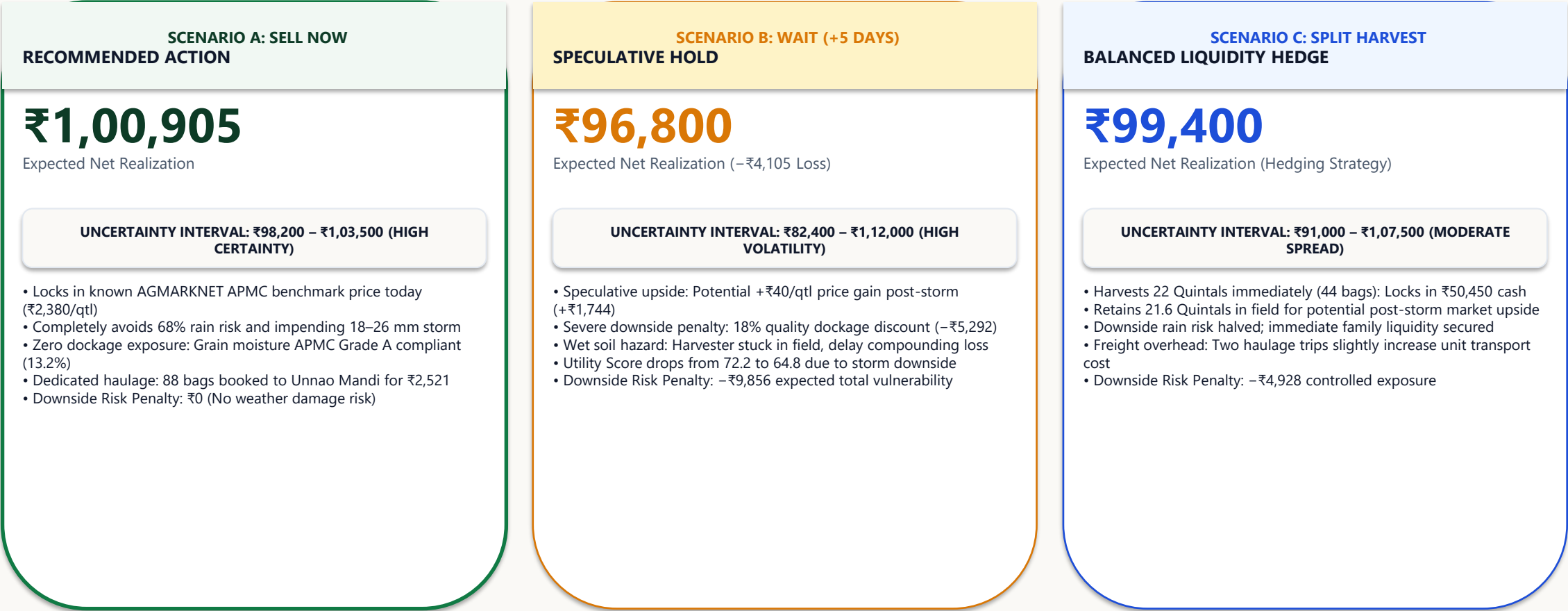
ONE FARM. ONE INTELLIGENCE LAYER.

Replacing 5 disconnected websites with a single coherent reality grounded in real authenticated farm state.



WHAT IF?

Before making an irreversible harvest decision, see what each choice could actually cost.



“We don’t pretend to know exactly what will happen. We show how the distribution of possible outcomes shifts with every choice — and keep the farmer in control.”

EVERY NUMBER HAS A STORY.

Destroying the AI black box with end-to-end data provenance, mathematical auditability, and truthful source labeling.

THE PROVENANCE CHAIN: DISPLAYED VALUE → FORMULA & MATH → SOURCE OF TRUTH → TIMESTAMP & FRESHNESS → CONFIDENCE SCORE

TC-WEATHER: WEATHER TELEMETRY

Open-Meteo Numerical NWP (ECMWF IFS & GFS Model Grid)

- Ingests 7-day numerical weather predictions
- Freshness window: 45 min cache invalidation
- Accurately labeled: Numerical model grid interpolation
- Absolute truth: Never called live radar or Doppler radar

TC-MARKET: APMC MARKET RATES

AGMARKNET / DMI Bulletin (Official Daily Modal Settlement)

- Mandi auction closing prices by Ministry of Agriculture
- Optimal yard selection: Unnao ₹2,380 vs Kanpur ₹2,310
- Accurately labeled: Official daily reference modal quote
- Absolute truth: Never claimed as a live stock ticker

TC-LOGISTICS: RURAL HAULAGE TARIFF

Geodesic Haversine Detour (1.25x Rural Curvature Factor)

- Great-circle distance calibrated with 1.25x detour factor
- Dedicated freight tariff: ₹2,000 base + ₹12/qtl weight scaling
- Freight to Unnao (28.4 km): ₹2,521 dedicated haul
- Absolute truth: Road distance estimate, never fake GPS

TC-AGRONOMY: BIOLOGICAL MATURITY

Deterministic GDD Engine (Thermal Units (T_base = 5.0°C))

- Cumulative GDD: $\sum \max((T_{\max} + T_{\min})/2 - 5.0, 0)$
- Biological maturity: 1,845 / 1,950 GDD (94.6% harvest-ready)
- APMC Grade A compliance: 13.2% grain moisture (Safe)
- Absolute truth: Agronomic physics, never static calendar days

TC-FORECAST: PRICE UNCERTAINTY

Quantile Horizon Estimator (Parameterized Mandi Baseline)

- Computes P10/P50/P90 price intervals using $\sqrt{h+1}$ spread
- Precipitation penalty: up to 18% moisture dockage discount
- Causal explainability: Bayesian risk aversion weighting
- Absolute truth: Quantile math, never fake neural networks

GRACEFUL DEGRADATION: If an external feed fails, KISAN COMPASS flags cached benchmarks, lowers confidence scores, and alerts the user — it NEVER fabricates fake data.

BUILT TO WORK IN THE REAL WORLD.

Engineered for immediate deployment across Indian agriculture without expensive hardware dependencies or black-box risk.

WHY IT IS FEASIBLE TODAY

Zero In-Field Hardware Dependency

Does not require expensive IoT soil sensors or weather stations. Operates on farmer-verified inputs and open satellite/NWP public models.

Lightweight & Low-Bandwidth Optimized

Built with React + Vite frontend and Supabase cloud. Consumes under 1.1MB total assets; loads reliably over rural 3G/4G connections.

Postgres Row-Level Security (RLS)

Multi-tenant cloud architecture with cryptographically enforced tenant isolation. Every farmer's farm and financial data is strictly private.

Human-in-the-Loop Safeguards

System provides explainable decision support rather than risky automated execution. Eliminates legal and operational liability.

WHAT CHANGES FOR THE INDIAN FARMER

BEFORE KISAN COMPASS (FRAGMENTED STATUS QUO)

- Checking 4–5 different apps with contradictory indicators
- Guessing which mandi yields more money without factoring freight
- Panic selling or holding blindly into convective storm damage
- Shocked by 15–20% moisture dockage deductions at the mandi yard
- Regret after irreversible harvest and transport decisions

AFTER KISAN COMPASS (DECISION INTELLIGENCE)

- One single unified digital twin representing the farmer's actual field
- Net realization calculated in rupees after dedicated freight tariffs
- Simulates counterfactuals before cutting the crop: Sell Now vs Hold vs Split
- Transparent downside dockage warning protects grain quality value
- High-confidence decisions backed by clear mathematical proof

SOCIO-ECONOMIC IMPACT: From information overload to decision clarity — preventing avoidable distress sales across rural India.

NOT ANOTHER FARM DASHBOARD.

Dashboards display data. KISAN COMPASS delivers contextual decision intelligence.

TRADITIONAL / FRAGMENTED FARM TOOLS

- **FRAGMENTATION:** Weather, mandi rates, and crop data in 4–5 separate apps
- **PRICING ILLUSION:** Displays raw commodity quotes ignoring freight tariffs & tolls
- **FAKE CERTAINTY:** Single-point forecasts pretending tomorrow's weather is 100% known
- **BLACK-BOX ADVICE:** Generic prescriptive commands with zero mathematical proof
- **FARMER BURDEN:** Leaves 100% of the cognitive synthesis and financial risk on the farmer

KISAN COMPASS: DECISION INTELLIGENCE

- **UNIFIED DIGITAL TWIN:** Persistent farm state bound to real field GPS & stand
- **TRUE NET REALIZATION:** Real cash income in rupees after dedicated rural haulage
- **WHAT-IF SIMULATION:** Evaluates Sell Now vs Wait vs Split trade-offs before acting
- **AUDITABLE TRUTH:** 12-step verifiable mathematical proof sequence for every claim
- **ABSOLUTE HUMAN CONTROL:** Autonomous execution disallowed; farmer holds total veto

THE REPRODUCIBLE PRODUCT LOOP: OBSERVE → UNDERSTAND → SIMULATE → DECIDE → EXECUTE → LEARN

"KISAN COMPASS doesn't decide for the farmer."

"Not what to think. What's actually true — and how sure we are."